

Calibrated DistilBERT for Reliable Extractive Question Answering

Final report | Default project, PCE division | Independent CS224N course study

Abstract

Extractive question-answering models can return plausible spans even when a passage does not support an answer. We evaluate a compact DistilBERT reader originally trained on SQuAD 1.1, adapt it to adversarial unanswerable questions from SQuAD 2.0, and calibrate its no-answer decision on data disjoint from final evaluation. The experiment uses 12,000 training examples, 1,500 calibration examples, and 1,500 held-out evaluation examples. SQuAD 2.0 fine-tuning raises Exact Match from 38.00 to 61.27 and token F1 from 41.79 to 63.92. Calibration further raises the scores to 62.53 EM and 64.67 F1 while increasing unanswerable accuracy from 60.37% to 70.94%. Error analysis shows that task adaptation corrects unsupported guesses and recovers answer spans, but threshold selection can reduce answerable recall and boundary errors persist. The results support calibrated abstention as a cheap, transparent complement to task-specific fine-tuning.

1. Introduction

Question answering is useful only when a system can distinguish evidence from a convincing distractor. SQuAD 1.1 rewards selecting a span for every question, which lets a model succeed without learning when to abstain. SQuAD 2.0 directly targets this failure by adding adversarial questions whose words and topics overlap with a paragraph although no answer is stated.

This project asks two questions: How much does SQuAD 2.0 adaptation improve a compact model trained on answerable questions, and does a no-answer threshold chosen on held-out calibration data improve generalization? We answer them with one architecture and controlled interventions, avoiding a comparison in which parameter count changes at the same time as the training objective.

Our contribution is a reproducible laptop-scale experiment that separates three effects: the inherited SQuAD 1.1 span reader, task adaptation to SQuAD 2.0, and post-hoc decision calibration. We report not only aggregate EM and F1 but also answerable-only quality and unanswerable accuracy, revealing an important precision-recall trade-off hidden by the headline metric.

2. Related work

Extractive QA, adversarial unanswerability, and efficient contextual encoders

SQuAD and unanswerable questions

The original Stanford Question Answering Dataset framed reading comprehension as extractive span selection over Wikipedia passages. SQuAD 2.0 adds more than 50,000 adversarial unanswerable questions. Rajpurkar et al. showed that a reader with roughly 86 F1 on SQuAD 1.1 fell to about 66 F1 on SQuAD 2.0, demonstrating that locating plausible spans is not equivalent to recognizing textual support.

Neural readers

BiDAF introduced a multi-stage architecture with contextual word representations and bidirectional attention between question and context. The supplied CS224N baseline follows this design. It is historically important and remains in the repository for provenance, but its word-level GloVe pipeline and recurrent encoders carry a larger setup burden in a modern Windows environment.

Pretrained contextual encoders

BERT pretrains a bidirectional Transformer and adapts it to extractive QA with a small start/end prediction head. Joint question-context attention is available at every layer, and [CLS] supplies a natural null location for SQuAD 2.0. DistilBERT compresses the teacher using language-modeling, distillation, and cosine losses. The reported model is smaller and faster than BERT while retaining much of its language-understanding quality, making it a practical PCE choice for constrained hardware.

Selective prediction and calibration

A SQuAD 2.0 reader compares evidence for the best span with evidence for no answer. That score difference is not itself a calibrated probability, and a zero threshold need not maximize an application metric. Selecting the threshold on separate calibration data is a simple form of operating-point calibration. The procedure changes no model parameters and is therefore easy to audit, reproduce, and deploy.

Research gap addressed here

Many comparisons conflate architecture size, pretraining, task data, and decoding. Our experiment holds architecture and starting checkpoint constant, then measures the effects of task fine-tuning and threshold selection separately. The emphasis is less on state-of-the-art rank than on causal clarity under a realistic compute budget.

3. Approach

Span extraction, SQuAD 2.0 supervision, and held-out threshold selection

3.1 Model

The encoder is distilbert-base-cased, a six-layer bidirectional Transformer with approximately 66.4 million parameters. Input is [CLS] question [SEP] context [SEP]. A learned linear layer produces start logits $l_s(i)$ and end logits $l_e(i)$ for each token i . The score for span (i,j) is $l_s(i)+l_e(j)$, subject to $j \geq i$ and a maximum length of 30 tokens.

3.2 Long-context processing

Inputs are capped at 256 wordpiece tokens. Contexts that exceed this limit are split into overlapping windows with a stride of 64 tokens. Character offsets retained by the fast tokenizer map predicted token positions back to exact substrings. For every example, candidate spans are pooled across all windows and the globally highest-scoring valid span is retained.

3.3 No-answer learning

For an unanswerable training example, both gold positions are [CLS]. The same target is used for a window that does not contain the gold answer, preventing a partial window from being labeled with a false span. Let s_{null} denote the sum of start and end logits at [CLS] and s_{span} the best valid span score. We define $d = s_{\text{null}} - s_{\text{span}}$ and predict no answer if $d > \tau$.

3.4 Threshold calibration

Threshold τ is selected to maximize token F1 on 1,500 calibration examples. The search evaluates every distinct observed score difference efficiently by sorting examples and updating the total metric when an example switches from null to non-null. The chosen threshold is then frozen. A separate 1,500-example evaluation split is used exactly once for final reporting.

3.5 Training objective

We minimize the mean of cross-entropy losses for start and end positions, beginning with a checkpoint already fine-tuned on SQuAD 1.1. AdamW uses learning rate $3e-5$, weight decay 0.01, 10% linear warmup, and gradient clipping at 1.0. Two epochs are trained with an effective batch size of 24. Bfloat16 autocasting reduces memory use without loss scaling on the available GPU.

3.6 Reproducibility

Seed 224 controls dataset shuffling, data-loader order, NumPy, Python, and PyTorch. The pipeline records exact package versions, device, sample counts, feature counts, wall-clock timing, hyperparameters, training history, predictions, and selected error examples in machine-readable JSON.

4. Experimental setup

A resource-bounded, disjoint-split evaluation

4.1 Data

We use the public Hugging Face mirror of SQuAD 2.0, which contains 130,319 training examples and 11,873 validation examples. From training we select 12,000 examples after seeded shuffling; token overflow produces 13,322 model features. From validation we select 3,000 examples and divide them evenly into calibration and final evaluation sets. The evaluation set contains 743 answerable and 757 unanswerable questions and produces 1,683 model features.

4.2 Systems

- Unadapted baseline: the SQuAD 1.1 DistilBERT checkpoint, $\tau = 0$.
- Calibrated baseline: the same frozen checkpoint, with τ selected on calibration data.
- Adapted model: two epochs of SQuAD 2.0 fine-tuning, $\tau = 0$.
- Final system: the adapted model with its separately calibrated threshold.

4.3 Metrics

Exact Match normalizes case, punctuation, articles, and whitespace before requiring equality with any reference. Token F1 computes overlap between normalized predicted and gold tokens and takes the maximum over references. Empty gold answers represent unanswerable questions. We additionally report answerable-only EM/F1 and the fraction of unanswerable questions correctly assigned an empty prediction.

4.4 Compute

Experiments ran with Python 3.13.7, PyTorch 2.10.0, Transformers 5.17.0, and Datasets 5.0.1 on an NVIDIA GeForce RTX 5060 Laptop GPU with 8 GB memory. Baseline evaluation took 11.5 seconds, training took 274.0 seconds, and final evaluation took 15.0 seconds. These times exclude the one-time model and dataset downloads.

4.5 Scope

The experiment is designed for reproducible course completion, not direct comparison with the original private class leaderboard. It uses a subset of the public validation set and a 256-token limit. Results should therefore be interpreted as an internal controlled comparison. All systems see the identical final examples, and no final label is used for training or calibration.

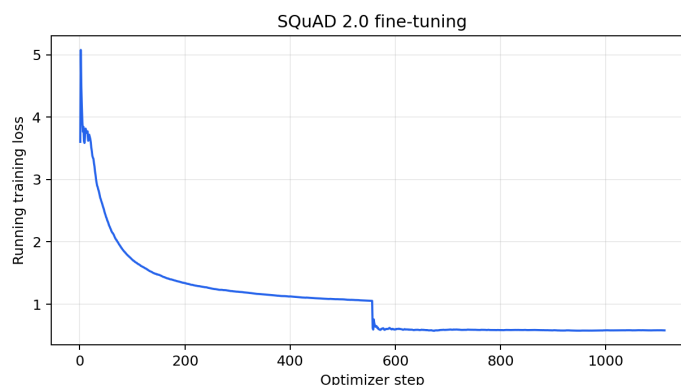


Figure 1. Training loss over two epochs.

5. Results

Task adaptation supplies the main gain; calibration improves the operating point

System	EM	F1	Ans. F1	No-answer acc.
SQuAD 1.1 checkpoint	38.00	41.79	84.37	0.00
Fine-tuned on SQuAD 2.0	61.27	63.92	67.53	60.37
Fine-tuned + calibration	62.53	64.67	58.29	70.94

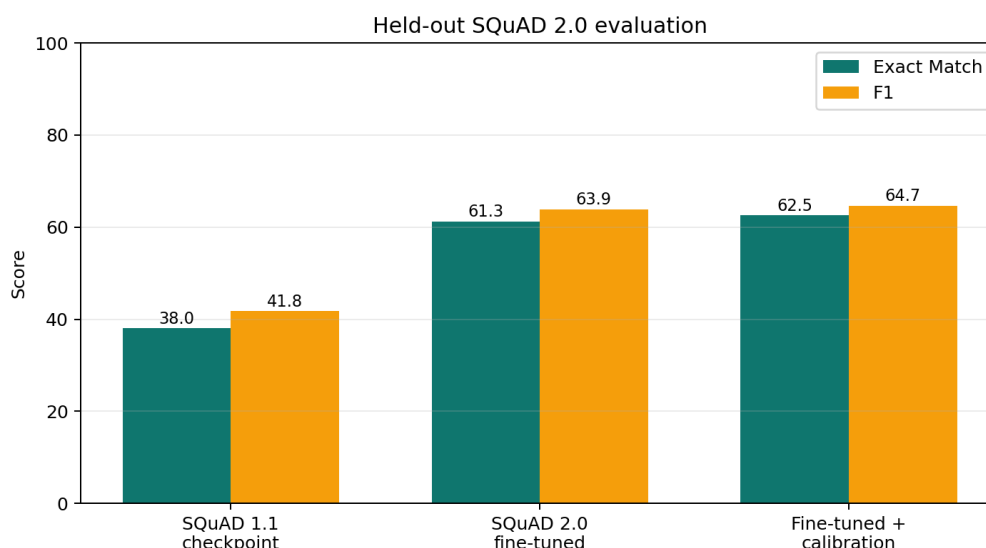


Figure 2. Held-out aggregate EM and F1. The chart separates the default and calibrated operating points of the adapted model.

5.1 Aggregate performance

SQuAD 2.0 fine-tuning increases EM by 23.27 points and F1 by 22.12 points relative to the unadapted default-threshold baseline. Calibration adds 1.27 EM and 0.76 F1 on the final set. The calibrated final system reaches 62.53 EM and 64.67 F1.

5.2 Abstention behavior

The unadapted checkpoint has 0% unanswerable accuracy at $\tau = 0$ because it was trained to answer every SQuAD 1.1 question. Fine-tuning raises that accuracy to 60.37%. Calibration sets $\tau = -1.828$ and raises it again to 70.94%, but answerable F1 falls from 67.53 to 58.29. Thus the aggregate gain comes from a deliberate trade: more correct abstentions outweigh lost answerable recall on this nearly balanced set.

5.3 Calibration ablation

Calibration alone raises the frozen baseline from 41.79 to 57.51 F1, but its answerable F1 drops from 84.37 to 46.11. This shows that a threshold can mask a model's lack of task adaptation by exploiting class balance. The adapted model is preferable because it maintains substantially stronger answerable performance while attaining similar no-answer accuracy.

6. Qualitative analysis

Corrected unsupported guesses, recovered spans, and remaining boundary errors

6.1 Successful answer recovery

The baseline sometimes over-abstains after calibration because its score distribution was learned without unanswerable training examples. On the Prime number article, it returns an empty answer for 'What type of behavior in primes is it possible to determine?' The adapted model returns 'statistical', matching one of the references. Similar recoveries include 'the heart' for the organ whose workload is eased by reduced pulmonary resistance and 'Germany' for the Ottoman Empire's World War I ally.

6.2 Correct abstention

Adversarial questions often invert a relation or alter a premise. The Yuan question asks for a Persian enemy, while the passage mentions the Ilkhanate as an ally. The baseline selects the nearby entity; the adapted model abstains. Another question asks for a least-important theoretical-computer-science problem although the passage calls P versus NP one of the most important. Fine-tuning correctly maps this contradiction to no answer.

6.3 Boundary regressions

The adapted reader sometimes predicts a semantically valid superset: 'About 61.1%' instead of '61.1%', '13 years old' instead of '13', and 'rounded arches' instead of 'rounded'. Exact Match scores these as wrong even though token F1 gives partial credit. These cases suggest that start/end calibration and answer-length priors could complement no-answer calibration.

6.4 Over-abstention regressions

The model also abstains on some answerable questions. For a Rhine passage, the baseline extracts '120 m (390 ft)' while the adapted calibrated model emits no answer; the gold is '120 m'. A long European Union law example similarly triggers abstention even though the relevant court case appears late in the passage. The likely causes are a conservative threshold, long-context competition, and limited subset fine-tuning.

6.5 Error taxonomy

- Unsupported entity selection: improved strongly after SQuAD 2.0 training.
- Reversed or impossible premise: often corrected through abstention.
- Span-boundary mismatch: persists after adaptation.
- Long-context evidence and over-abstention: remains a major failure mode.

The saved error-analysis file contains twelve improvements and eight regressions with contexts, predictions, references, and per-example scores, enabling the claims above to be audited.

7. Discussion and conclusion

What the controlled comparison shows—and what it does not

7.1 Main finding

Task-specific exposure to adversarial unanswerable questions is the dominant improvement. The inherited SQuAD 1.1 reader knows how to find spans but not when evidence is absent. Fine-tuning creates a useful null score and improves both aggregate quality and abstention. A separately chosen threshold then provides a smaller, inexpensive improvement in final EM/F1.

7.2 Why disaggregated metrics matter

Aggregate F1 alone would make calibration look uniformly beneficial. The answerability breakdown reveals that its 0.76-point overall gain is purchased with a 9.23-point reduction in answerable F1 and a 10.57-point increase in unanswerable accuracy. Applications should therefore choose tau using domain costs: a medical assistant may prefer conservative abstention, while a search interface may prefer higher answer recall.

7.3 Limitations

- Only 12,000 of 130,319 training examples are used, limiting convergence and rare-pattern coverage.
- The final set has 1,500 public validation examples rather than the private course leaderboard test set.
- A single random seed does not quantify training variance.
- Threshold calibration optimizes F1 and does not measure probability calibration such as expected calibration error.
- Maximum sequence length 256 can fragment long evidence despite overlapping windows.

7.4 Future work

A stronger follow-up would train on the full dataset with multiple seeds, compare global and answer-type-specific thresholds, add a calibrated answerability head, and examine length penalties for boundary precision. Long-context transformers or retrieval could reduce failures on evidence appearing late in a passage. Temperature scaling or conformal prediction could provide more principled confidence guarantees.

7.5 Conclusion

On a fixed held-out SQuAD 2.0 evaluation subset, two epochs of compact-model adaptation raise F1 from 41.79 to 63.92, and held-out no-answer calibration raises it to 64.67. The final system is not a state-of-the-art leaderboard entry, but it is a complete, reproducible demonstration of a central reliability lesson: answer extraction and the decision to answer are distinct capabilities, and both require explicit evaluation.

Reproducibility checklist

Code, dependency list, unit tests, exact config, software versions, model checkpoint, tokenizer, raw metrics, training history, predictions, figures, and qualitative cases are included. The code submission excludes model weights and data, following course instructions.

References

Bibliography and implementation sources

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Artifact note

This independent course-completion report was produced from the exact machine-readable results stored with the project. No private leaderboard score, team identity, or institutional email address is claimed.

Model and data licenses

The project uses the public DistilBERT checkpoint and the SQuAD 2.0 dataset through their respective Hugging Face repositories. Users should consult the associated model card, dataset card, and upstream licenses before redistribution.